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**Statement of integrity:** By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an “X” above).

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**Note:** You may be required to provide proof of your outreach to non-contributing members upon request.

N/A

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## Main Summary & Project Scope

The main task here is to use market data for option prices for SM stock , estimate model inputs from the given prices and then using the model to price the products as requested by the client. SM stock is trading at \$232.90. I used yearly risk free rate of 1.50% and assumed 250 trading days in the trading year.

For short term 15-day option- Heston[1] model is something which give a some reasonable fit to the market trading prices , with root mean square of 0.4954 and mean average error of 0.3725. But however some the estimated parameter ends up at the edge of the allowed ranged. Because of all this- my recommendation would be to treat these results with some caution than assuming the model fit perfect.

I used the calibrated Heston[1] model, to price the 20-day at the money Asia call option. The fair price was obtained as \$4.7841 per option unit. Then I added the 4% bank fee , the final price charged to the client come to be \$4.9755 per option unit

For the 60 day option, I pivoted to use the Bates[2] model and calibrated using both the Lewis and Carr-Madan approaches . The output from both these two methods very similar pricing errors , and obtained RMSE value was close to 1.155 . The suggestion which option prices from these two method gave broadly consistent results for 60 day option data

For the interest free rate part –CIR model was fitted to the Euribor curve. When I ran simple simulation approach- the expected 12-moth Euribor rate after 1 year comes to 5.4788% . The 95% simulated range from lies 3.9702% to 7.2013% . This implies that the model points to very high interest rates over the next year, but the result should be viewed as model estimate rather being a guaranteed forecast ones.

## Data and Basic Setup

I am using both calls and puts together as both types give useful information about the market prices.

15-day used for short term Heston work.

60 day used for the Bates[2] model.

Each slice has 10 options: 5 calls and 5 puts

**Table 1: Option Data used in the report**

| Model Stage                | Target Maturity | Selected Maturity | Options Used | Call Options | Put Options |
|----------------------------|-----------------|-------------------|--------------|--------------|-------------|
| Step 1: Heston Calibration | 15 Trading Days | 15 Trading Days   | 10           | 5            | 5           |
| Step 2: Bates Calibration  | 60 Trading Days | 60 Trading Days   | 10           | 5            | 5           |

For put option- put-call parity where I turn a call price into matching put price using the strike, stock price, time and interest rate. This keep pricing process consistent

## Step 1A – Heston Model for the 15-Day Options

**Table 2: Heston fitted values**

| Parameter                           | Value     |
|-------------------------------------|-----------|
| Mean Reversion Speed ( $\kappa$ )   | 0.797155  |
| Long-Run Variance ( $\theta$ )      | 0.000100  |
| Volatility of Variance ( $\sigma$ ) | 1.223516  |
| Correlation ( $\rho$ )              | -0.999000 |
| Initial Variance ( $v_0$ )          | 0.103581  |

**Table 3: Heston fit quality**

| Calibration Item               | Value                |
|--------------------------------|----------------------|
| Model                          | Heston               |
| Pricing Method                 | Lewis                |
| Target Maturity                | 15 Trading Days      |
| Selected Maturity              | 15 Trading Days      |
| Number of Options Used         | 10 (5 Calls, 5 Puts) |
| Mean Squared Error (MSE)       | 0.245427             |
| Root Mean Squared Error (RMSE) | 0.495406             |
| Mean Absolute Error (MAE)      | 0.372506             |
| Feller Condition               | Not Satisfied        |
| Optimizer Status               | Converged            |

The model as seen above converged, meaning that fitting prices finished. But however, two fitted values came as the edge of the allowed range. This is kind of warning- i.e. short 15 day option prices are hard for plain Heston model to match exactly perfectly. Still using this for Asian option prices because the required short term model- but I am treating fitted values carefully.

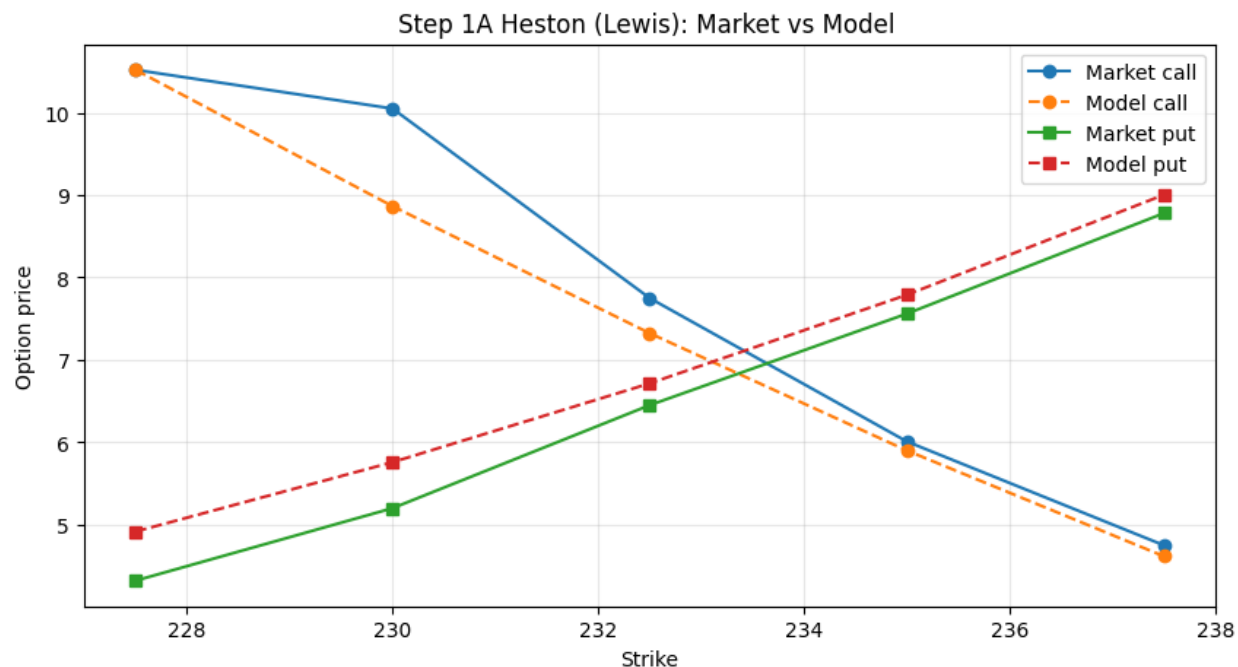


Figure 1: Heston model prices compared with market prices

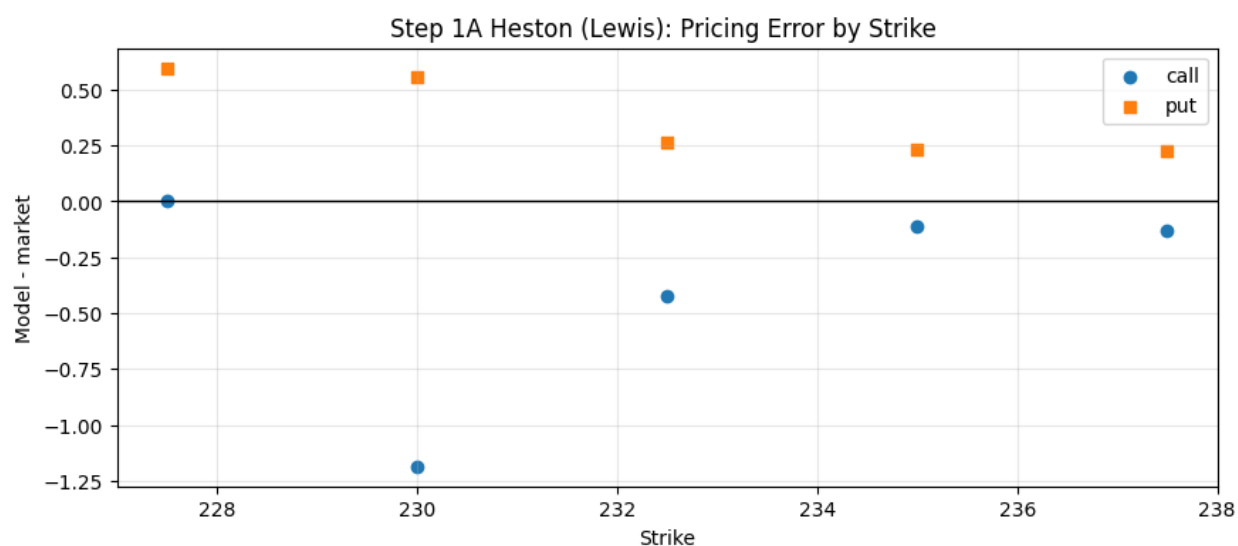


Figure 2: Heston pricing errors by strike

## Step 1C – 20-Day Asian Call Price

Table 4: Asian option price

| Pricing Item                  | Result               |
|-------------------------------|----------------------|
| Current Stock Price ( $S_0$ ) | 232.90               |
| Strike Price ( $K$ )          | 232.90               |
| Time to Maturity              | 20 Trading Days      |
| Number of Simulations         | 100,000              |
| Estimated Fair Price          | 4.784112             |
| Standard Error                | 0.017930             |
| 95% Confidence Interval       | 4.748969 to 4.819254 |
| Bank Fee (4%)                 | 0.191364             |
| Final Client Price            | 4.975476             |

I simulated 100,000 possible future paths for the SM stock prices. For each of the paths, mean is taken including the today's price, then payoff is calculated. After that we averaged all simulated payoffs and discounted to the present time. The fair price comes to \$4.7841. Required 4% fee is 0.1914. there final price comes to be \$4.9755

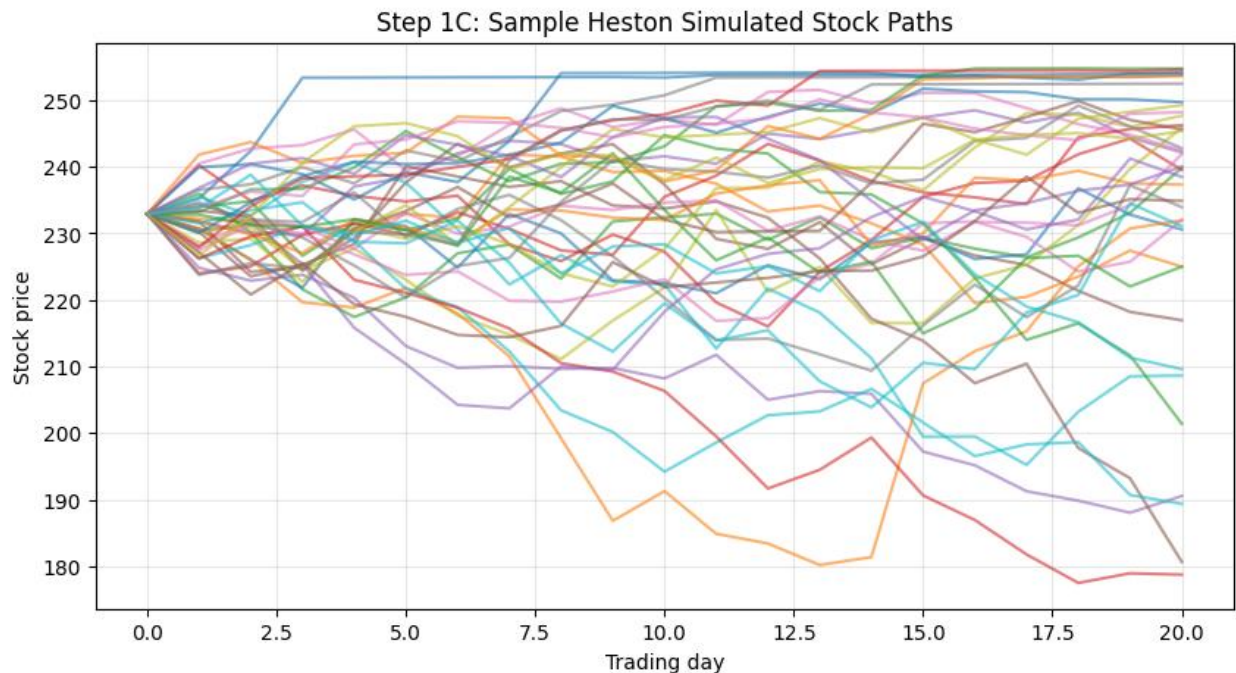


Figure 3: Example simulated SM stock paths

## Step 2 – Bates Model for the 60 Day Options

As client considering a longer maturity of 60 trading days-I then pivot to Bates model. Bates [2] model is like the Heston model, but it however allows for sudden jumps in the stock price. This helps to show jump risk.

I fit the Bates model by two methods: One by Lewis[4] and other by Carr-Madan. Both are recommended pricing methods.

**Table 5: Bates fitted values**

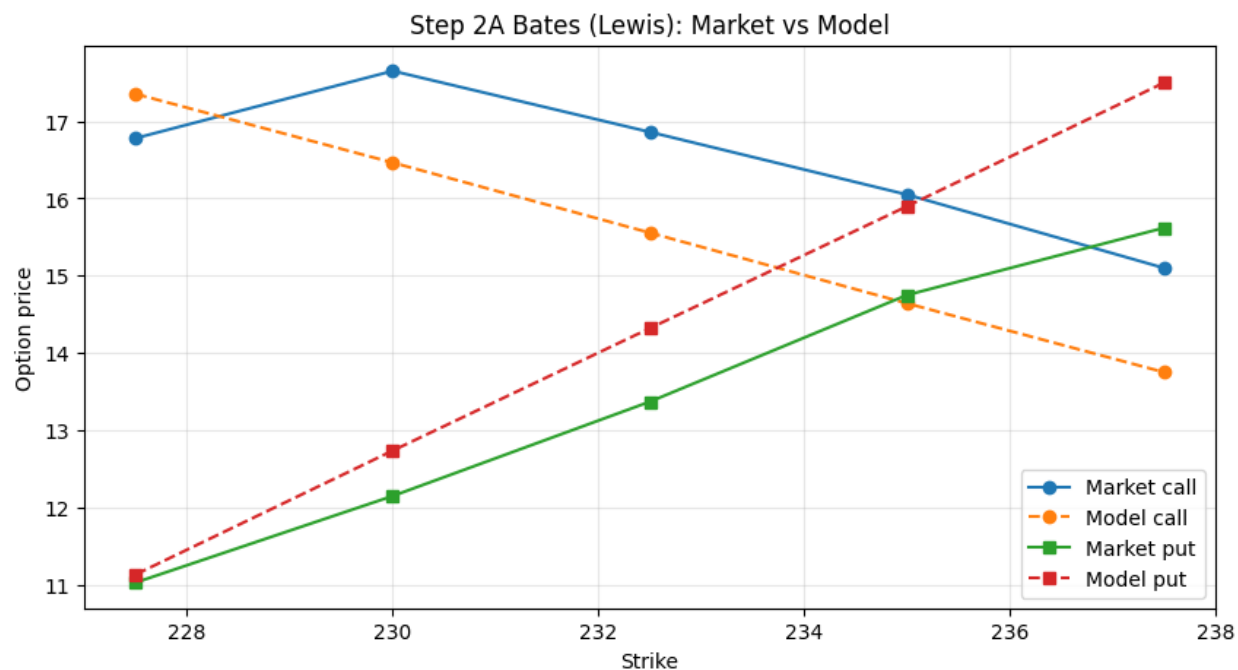
| Parameter                             | Lewis Method  | Carr–Madan Method |
|---------------------------------------|---------------|-------------------|
| Mean Reversion Speed ( $\kappa$ )     | 2.8871        | 3.0446            |
| Long-Run Variance ( $\theta$ )        | 0.003026      | 0.000100          |
| Volatility of Variance ( $\sigma_v$ ) | 0.2637        | 0.1498            |
| Correlation ( $\rho$ )                | -0.9990       | -0.9990           |
| Initial Variance ( $v_0$ )            | 0.000100      | 0.001182          |
| Jump Intensity ( $\lambda_j$ )        | 1.8736        | 1.8762            |
| Mean Jump Size ( $\mu_j$ )            | 0.2138        | 0.2137            |
| Jump Volatility ( $\sigma_j$ )        | 0.0010        | 0.0010            |
| Feller Condition                      | Not Satisfied | Not Satisfied     |

**Table 6: Bates fit quality**

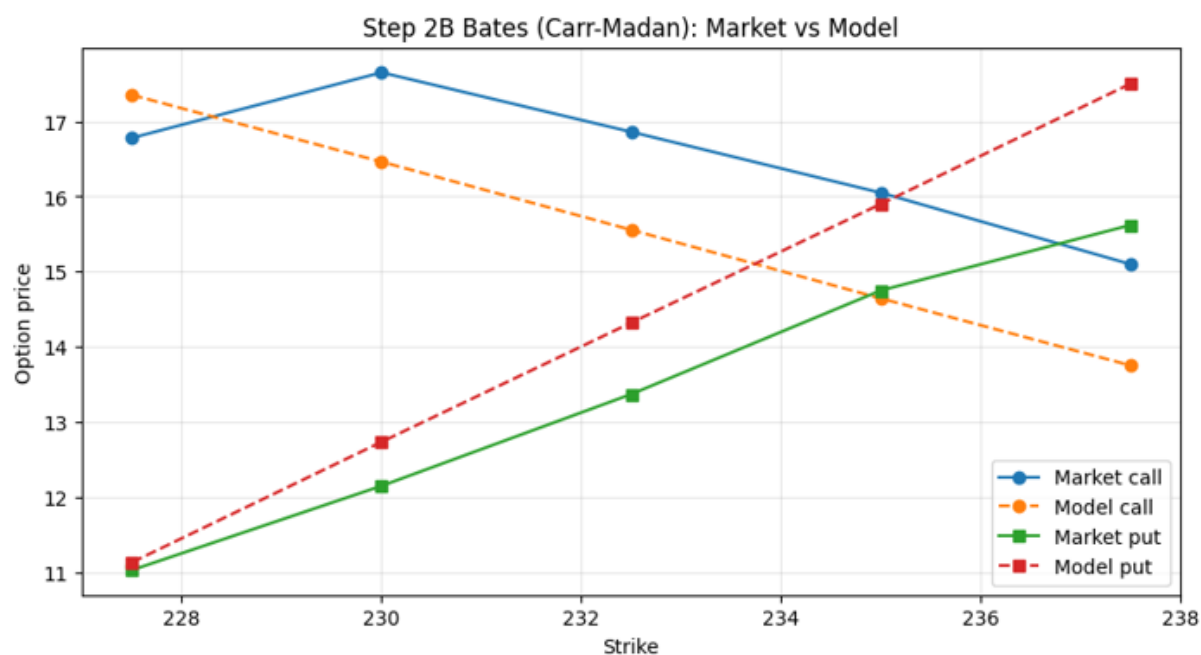
| Metric                         | Lewis Method | Carr–Madan Method |
|--------------------------------|--------------|-------------------|
| Target Maturity                | 60 days      | 60 days           |
| Selected Maturity              | 60 days      | 60 days           |
| Number of Options Used         | 10           | 10                |
| Mean Squared Error (MSE)       | 1.334112     | 1.333938          |
| Root Mean Squared Error (RMSE) | 1.155037     | 1.154962          |
| Mean Absolute Error (MAE)      | 1.047642     | 1.048408          |
| Optimizer Status               | Converged    | Converged         |

The Bates fits as seen above are very close to each other. Lewis has RMSE of 1.155037 and Carr Madan has RMSE of 1.1154962. Carr-Method is lower but the difference is very marginal small. AS we are using

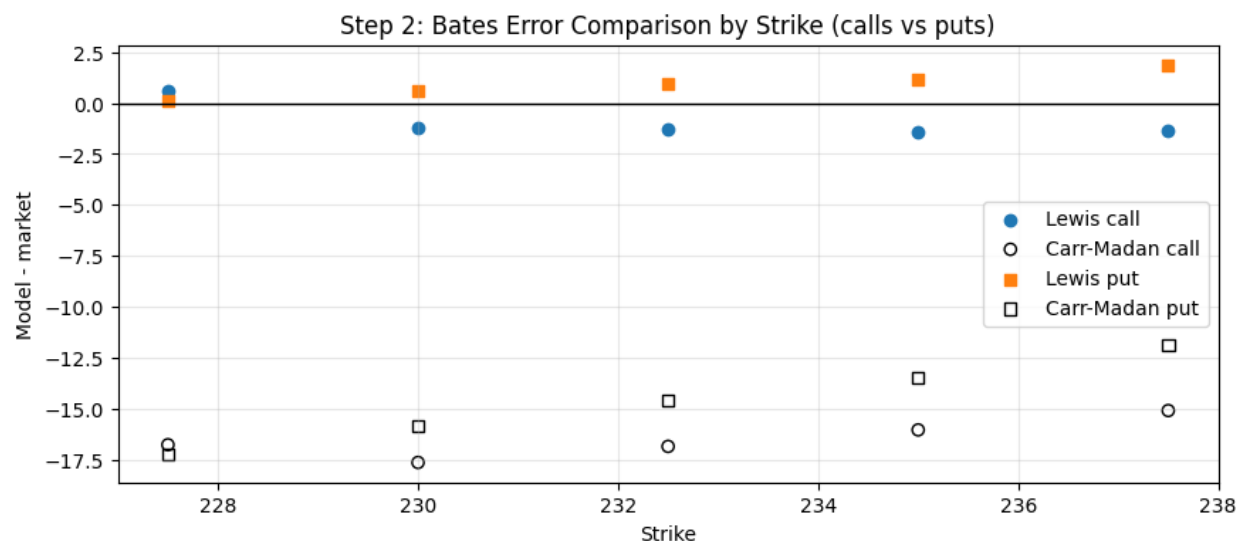
10 option only , these jump could not be read as forecast. This is small set of option prices for helping the model.



**Figure 5: Bates Lewis prices compared with market prices**



**Figure 6: Bates Carr-Madan prices compared with market prices**



**Figure 7: Bates pricing errors for Lewis and Carr- Madan**

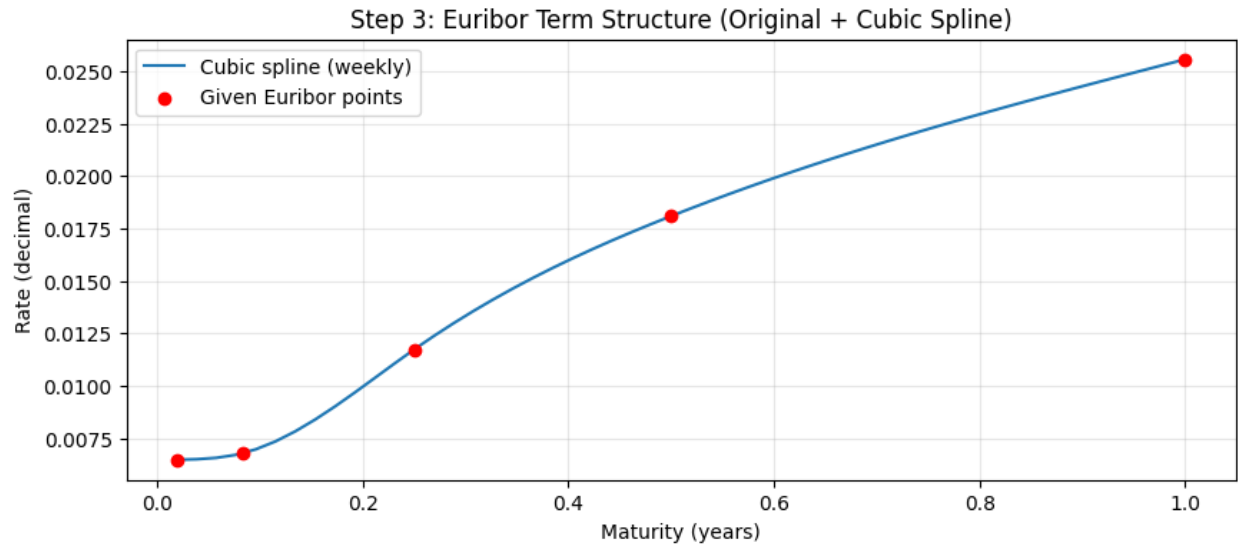
## Step 3 – Euribor Curve and CIR Rate Model

Last part of the method studies the interest rates. Here we use the Euribor rates given in project. Rate is showing how market prices money for different time periods i.e. from 1 week to 12 months.

We try to fill the missing rates between the given period using a smooth curve. Then I fitted the CIR model to that curve. CIR is useful as it prevents rates from becoming negative in normal cases

**Table 7: Euribor rates used**

| Tenor     | Maturity (Years) | Market Rate (%) |
|-----------|------------------|-----------------|
| 1 Week    | 0.0192           | 0.648           |
| 1 Month   | 0.0833           | 0.679           |
| 3 Months  | 0.2500           | 1.173           |
| 6 Months  | 0.5000           | 1.809           |
| 12 Months | 1.0000           | 2.556           |

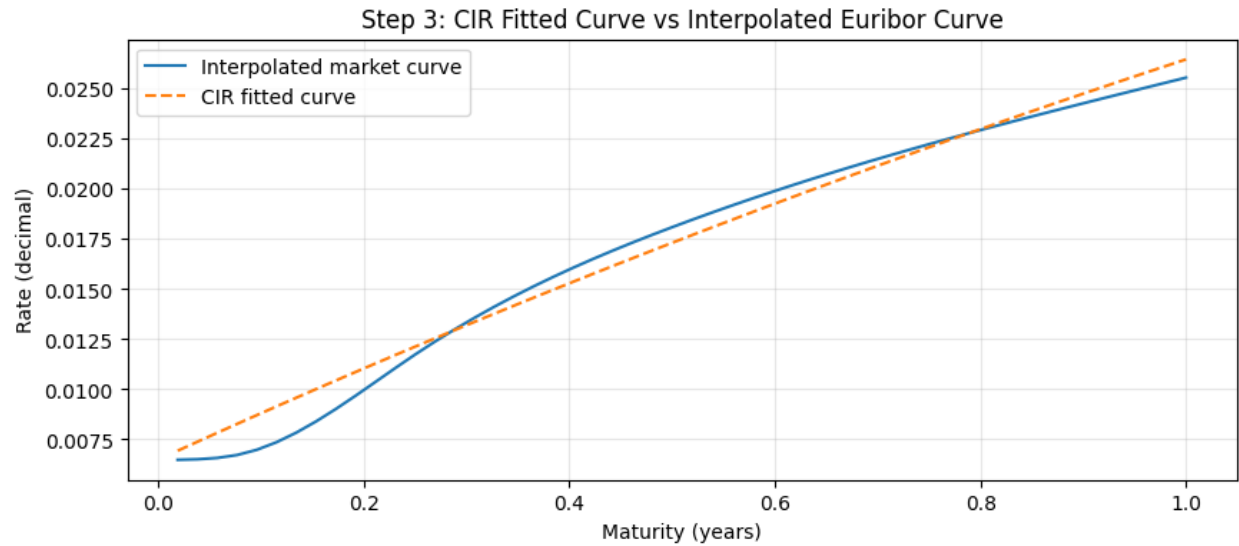


**Figure 8: Given Euribor points and smooth weekly curve**

**Table 8: CIR fitted values**

| Parameter / Metric                    | Result                |
|---------------------------------------|-----------------------|
| <b>r0</b>                             | 0.00648               |
| <b>Mean Reversion Speed (kappa)</b>   | 0.5122                |
| <b>Long-Term Mean Rate (theta)</b>    | 0.098438              |
| <b>Volatility (sigma)</b>             | 0.049952              |
| <b>Mean Squared Error (MSE)</b>       | $6.47 \times 10^{-7}$ |
| <b>Root Mean Squared Error (RMSE)</b> | 0.000805              |
| <b>Feller Condition</b>               | Passed                |

The CIR model fit the weekly Euribor curve quite closely to actual. RMSE is 0.000805 i.e. 8.05 basis points. Feller condition is also try which provides useful sigh that fitted process will behave correctly and reasonably.



**Figure 9: CIR-fitted curve compared with the Euribor curve**

## Step 3 – One-Year Euribor Simulation

Now, I will simulate the 12-month Euribor rates for the 12-month using the Monte Carlo paths of 100,000 paths. I simulated directly with CIR model, starting from the current 12-months Euribor for the whole interest rate curve.

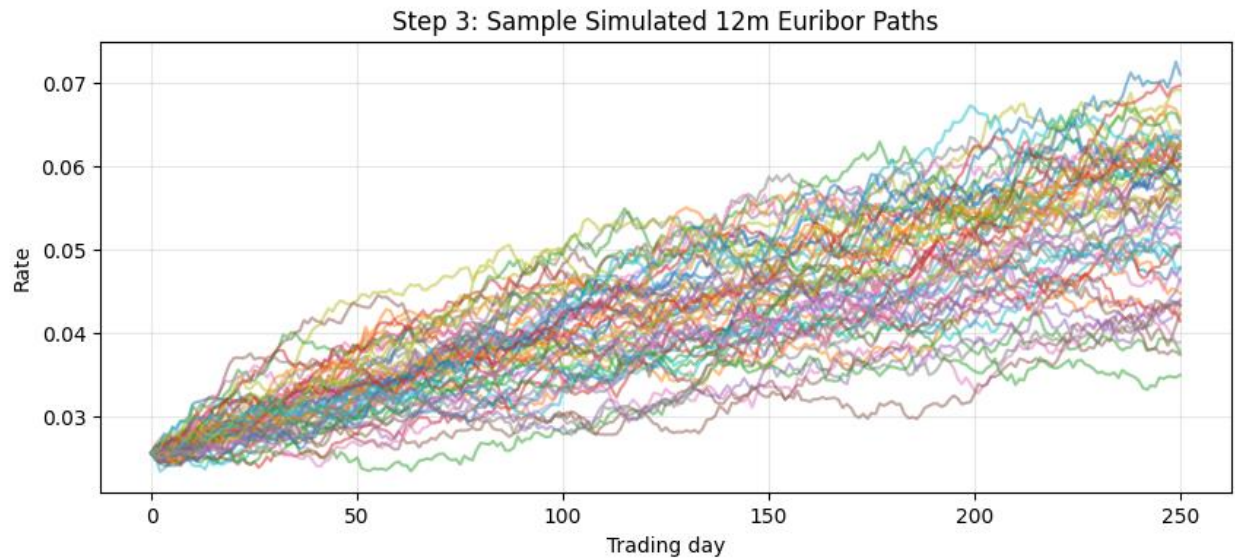
**Table 9: Simulated 12-months Euribor results**

| Simulation Item                     | Result  |
|-------------------------------------|---------|
| Current 12M Euribor                 | 2.56%   |
| Expected 12M Euribor (1 Year Ahead) | 5.48%   |
| Confidence Level                    | 95%     |
| Lower Percentile (95% CI)           | 3.97%   |
| Upper Percentile (95% CI)           | 7.20%   |
| Simulated Minimum                   | 2.14%   |
| Simulated Maximum                   | 9.50%   |
| Number of Simulations               | 100,000 |

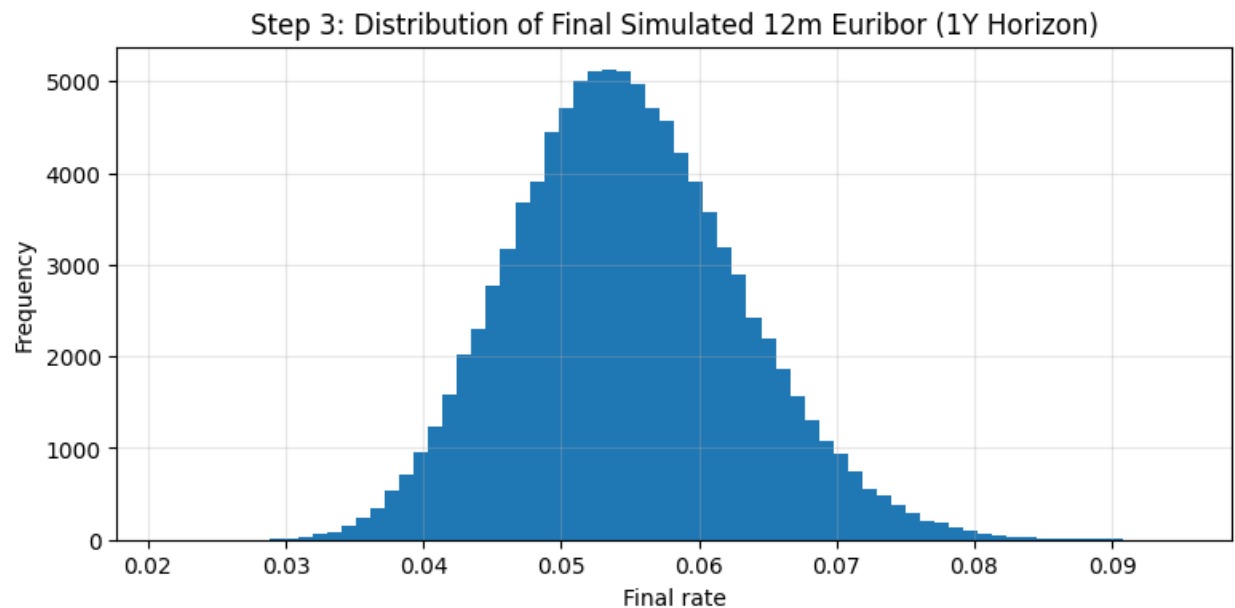
Expected 12-moth Euribor after 12 months is 5.4788%. 95% range is in 3.9702% to 7.2013%. This rainges means that the most of the simulated outcomes lands in these between two numbers.

If rates are higher later, then option prices can change. Higher rates tend to push the forward price higher which helps call but hurt put options, when all other conditions are the same. However but,

higher rates also increase discounting, lowering future cash flows when discounted to today. Hence desk should update the model before quoting future trades as these effects work in different directions.



**Figure 10: Example simulated 12-month Euribor paths**



**Figure 11: Distribution of 12-month Euribor after one year**

## Final Summary

- Fair value of the Asia option is ~\$4.7841
- Considering 4% bank fee- client price will be \$4.9755
- Though we used Heston mode for 15-day we found for 15-day option data- fitted values are on boundaries- hence it is very very important to treat result carefully
- The Bates model which we used for 60-day option data- we found that Lewis and Carr-Madan truns out to give almost same fit quality output.
- The simulation when we run for for the interest rate scenario suggest that higher when 12-month Euribor is considered under simple fitted model
- Finally- recommendation would be that- we rerun the full process under newest data before placing real trade

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